مدارس الإمارات الوطنية
Emirates National Schools

Design Department Weekly Lesson Plan

Dates: Oct 26 – 30, 2026 (Term 1 · Week 9) Grade/Class: MYP Year 3 (Grade 8) Teacher: Tichaona Gozo

Unit Title	Lesson Title	
AI-Assisted Product Prototyping & Campaign Design – Developing Ideas (Criterion B)	**Lesson 1 (W9.1): Using AI Tools to Refine & Visualise the Chosen Concept** **Lesson 2 (W9.2): Comparing & Justifying the Chosen Design (Sketch vs AI Refinement)**	
Learning Targets	**Success Criteria**	**Assessment tools and strategies** Formative / Summative
Lesson 1 (W9.1) – Students will be able to: • Write a structured prompt (subject, features, materials/colours, style, view, constraints) from their W8.2 hand-drawn sketch. • Generate at least two AI-refined iterations of their concept and record each prompt, output and change in a Prompt Log. • Compare the AI output with their original sketch and identify at least one change they would make. **Lesson 2 (W9.2) – Students will be able to:** • Compare the hand-drawn sketch and the AI-refined version against at least 3 design specifications. • State which version (or combination) they will take forward and justify it using at least one specification. • Explain one human design decision the AI output did not capture.	**Lesson 1** • **MUST:** I can generate one AI-refined visual from my sketch. • **SHOULD:** I can explain the prompt(s) I used and why. • **COULD:** I can compare the AI output to my original sketch and identify what I would change. **Lesson 2** • **MUST:** I can state which version (or combination) I am taking forward. • **SHOULD:** I can justify my choice using at least one specification. • **COULD:** I can explain a design decision I made that the AI output did not capture.	**Lesson 1 – Formative 2 of 2 (Criterion B)** AI-Refined Concept & Prompt Log – teacher feedback against Criterion B strands ii–iii (unweighted; feeds the Planning Drawings summative in W10.1). **Lesson 2 – Ongoing formative** Teacher observation, comparison-table draft check and peer feedback (unweighted). Exit tickets in both lessons.
Standards	**ATL Skills**	**National Identity Links**
MYP Criterion B (Developing Ideas): strands ii–iii – a range of feasible ideas (hand-drawn and AI-refined) and a chosen design justified with the specification. **CS.AP.1** Design and iteratively develop computational solutions using algorithms (draft → AI → refine).	• **Thinking – creative:** turning a sketch into precise prompts and iterations • **Thinking – critical:** judging AI output against specifications; keeping creative control	Students refine a product that solves a real UAE problem for the UAE Future Innovation Showcase, using AI as a tool while staying in charge of the design decisions. Technology, innovation, artificial intelligence and ethical digital design support the UAE's vision for a secure,

CS.CS.1 Describe how computing systems process, store and transmit information (how AI generators turn prompts into images) – L1. **Big idea:** *Creative control, informed by research and ethical AI use, enables designers to develop innovative products and campaigns that respond to the needs of UAE users.* **Key concept:** Development · **Related:** Innovation, Form, Function · **Global context:** Scientific & technical innovation – adaptation, ingenuity and progress	• **Media literacy:** responsible, transparent use of AI tools • **Communication:** justifying design decisions; giving peer feedback **GRASPS link:** as Product Designer & Creative Director, students produce the hand-drawn concept and the AI-refined visual prototype for the UAE Future Innovation Showcase.	innovative, knowledge-based economy; responsible digital citizenship, critical thinking and future-ready skills. **Cross-curricular:** ICT, Art & Design

Technology being used – physical, websites, Learning Support Platforms, Apps, etc.
Own devices, a school-approved AI image generator (e.g. Canva Magic Media / Adobe Express), School AI (LLM for prompt support), Toddle, Mentimeter, Lesson 1 & 2 PowerPoints, Prompt Log and comparison worksheets, students' W8.2 hand-drawn sketches.

Main Teaching and Learning Activities – Lesson 1 (W9.1): Using AI Tools to Refine & Visualise the Chosen Concept

Bellringer	**Spot the Prompt** *(pairs)* The teacher shows two AI images of the same product: one made from a vague prompt ("a water bottle"), one from a detailed prompt. Students match each image to its prompt and list 3 words that made the difference. Discussion: "Who made the design decisions – the designer or the AI?"
Time	5 min
Main Activities	**Mini-lesson – Prompt engineering for designers** (10 mins) The teacher introduces a prompt formula: Subject + Key features + Materials/Colours + Style + View/Background + Constraints. Live demo: the teacher converts an annotated hand sketch into a prompt, generates an image and improves the prompt once (iteration). Rule: the AI refines YOUR design and does not replace it. **Task 1 – Write my prompt** (10 mins) Using their annotated W8.2 sketch and design specifications, students write Prompt v1 with the formula. A partner checks it: does every key feature in the sketch appear in the prompt? **Task 2 – Generate and iterate** (15 mins) Students generate at least 2 iterations with the school-approved tool. In the Prompt Log they record each prompt version, the output (screenshot), what changed and why. The teacher circulates to check responsible use and to make sure students, not the AI, are making the decisions. **Task 3 – Designer control check** (10 mins) Students compare the AI visual with the original sketch: what the AI kept, changed and got wrong, and one change they would make (COULD). They upload the sketch, AI visuals and Prompt Log to Toddle.

Time	45 min
Plenary	**Exit Ticket (Mentimeter)** • "My most effective prompt change was ___ because ___." • "One design decision the AI did NOT make for me was ___."
Time	5 min
Resources	Lesson 1 PowerPoint (W9.1), Prompt Log worksheet, prompt-formula card, W8.2 hand-drawn sketches and design specifications, devices, AI image generator, Toddle, Mentimeter.
Main Teaching and Learning Activities – Lesson 2 (W9.2): Comparing & Justifying the Chosen Design	
Bellringer	**Debatable question – Stand on the line** "Does using AI make a product design more innovative, or can it reduce a designer's originality and responsibility?" Students stand on an agree/disagree line and 2–3 students justify their position.
Time	5 min
Main Activities	**Mini-lesson – Justifying with specifications** (8 mins) The teacher models the frame "Specification → Evidence → Decision": "Specification 2 says ___; the AI version shows ___ while my sketch shows ___; so I will ___." Students see a strong and a weak justification and say what makes the difference. **Task 1 – Side-by-side comparison** (15 mins) Students complete a comparison table rating the hand-drawn sketch and the AI-refined version against at least 3 design specifications (✓ / ◐ / ✗), with evidence for each rating. **Task 2 – Decide and justify** (12 mins) Students state which version, or combination, they will take forward (MUST) and justify it with at least one specification (SHOULD). They then explain one human design decision the AI did not capture (COULD). **Task 3 – Gallery walk peer feedback** (10 mins) Students display their sketch, AI visual and justification. Peers leave two stars and a wish on sticky notes, focusing on whether the justification uses the specifications.

Time	45 min
Plenary	**Exit Ticket** "I am taking forward ___ because specification ___ … One improvement I will make before my planning drawings (W10.1) is ___."
Time	5 min
Resources	Lesson 2 PowerPoint (W9.2), comparison & justification worksheet, Lesson 1 Prompt Logs and AI visuals, sticky notes, devices, Toddle.

Differentiation	
Support	• **L1:** fill-in-the-blank prompt-formula card, design adjective word bank and an example prompt; one required iteration instead of two. • **L2:** partially completed comparison table and sentence starters for the justification.
Challenge	• **L1:** use negative prompts / constraints and compare outputs from two AI tools, explaining the differences. • **L2:** justify using two or more specifications plus research evidence; evaluate the ethics of using AI-generated visuals in a campaign.
IEP	As per individual IEP accommodations (extra time, reduced written output, assistive technology, chunked instructions).
ILP	• Visual prompt cards and a vocabulary list (prompt, iteration, specification, justify, refine). • Paired work with a supportive peer; oral justification accepted in place of a written one.

Gifted & Talented	• **L1:** create two AI variants aimed at two different UAE user groups (e.g. residents vs tourists) and explain the design choices. • **L2:** draft a short design rationale that will open the W10.1 planning drawings.

Homework
• **Lesson 1:** Upload the sketch, AI-refined visuals and Prompt Log to Toddle. • **Lesson 2:** Act on peer feedback and finalise your justification ready for the Criterion B planning drawings summative (W10.1).
Reflection Notes
Lesson 1 (W9.1): **Lesson 2 (W9.2):**